laws of quantum mechanics. In particular, like all physical quantities, volume may not assume arbitrary values but only certain particular ones, as I described in Chapter 4. The list of all possible values is called, if you remember, the spectrum. Hence there should exist a ‘spectrum of the volume’ (figure 6.2)

Dirac provided us with the formula with which to compute the spectrum of every variable. The calculation took time, first to formulate it and then to complete it, and made us suffer. It was completed in the mid-1990s, and the answer, as expected (Feynman used to say that we should never do a calculation without first knowing the result), is that the spectrum of the volume is discrete. That is, the volume can only be made up of ‘discrete packets’. These are somewhat similar to the energy of the electromagnetic field, which is also formed of discrete packets: photons.

The nodes of the graph represent the discrete packets of volume and, as in the case of photons, can only have certain sizes, which can be computed using Dirac’s general quantum equation.[fn34] Every node n in the graph has its own volume v_n: one of the numbers in the spectrum of the volume. The nodes are the elementary quanta of which physical space is made. Every node of the graph is a ‘quantum particle of space’. The structure that emerges is the one illustrated in figure 6.3.

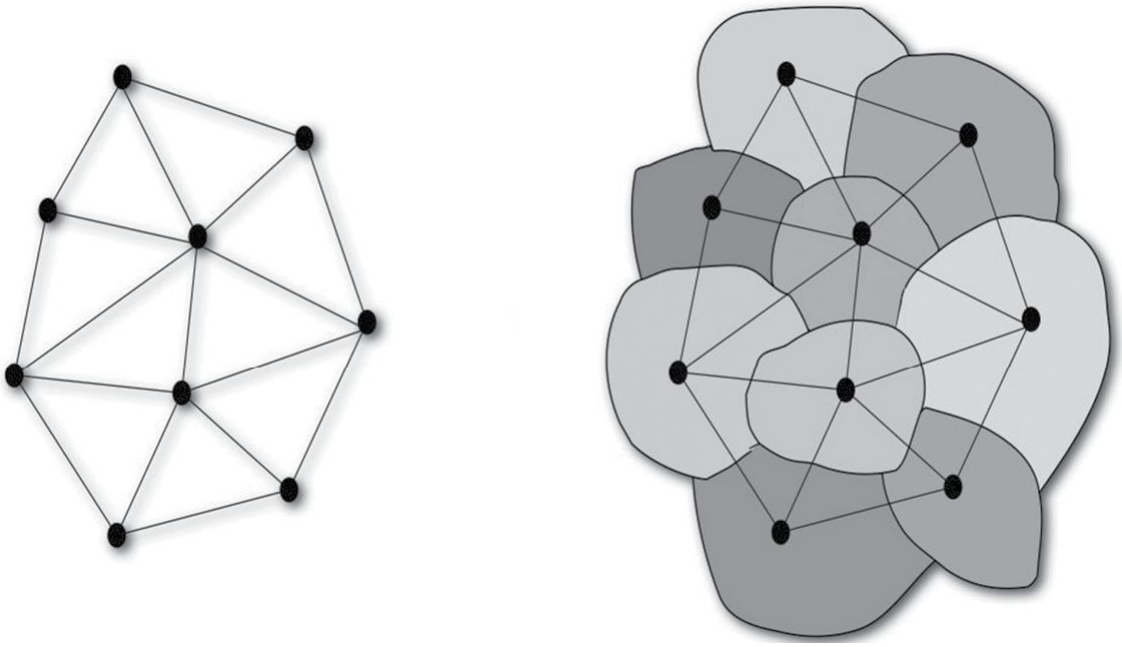


Figure 6.3 On the left, a graph formed by nodes connected by links. On the right, the grains of space which the graph represents. The links indicate the adjacent particles, separated by surfaces.

A link is an individual quantum of a Faraday line. Now we can understand what it represents: if you imagine two nodes as two small ‘regions of space’,